Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

What will help me through class will be lists, dataclasses, for loops, and 2d lists. I am hoping to develop my skills of creating better dataclasses and being able to incorporate drafter better in my programing skill library

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I feel the easiest part is going to be the functions because when I do functions I feel I take an approach that specifically makes things easier. The hardest part is going to be using drafter and specifically working with creating pages because that is the hardest for me to do.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

The tasks that will take the most work are going to be the pages and drafter route functions. I will implement the home page first to be able to make that be the starting point where I can go on from there. Second will be creating similar functions like the ability to convert the aft events into scores.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

What will interfere with my work the most is going to be my problem of completing something until 90% then stopping until right at the deadline. The resources I will use are my friends for me to look up to them, seeing them actually doing the work which will then have me do the work. I do not have any questions about the assignment requirements.

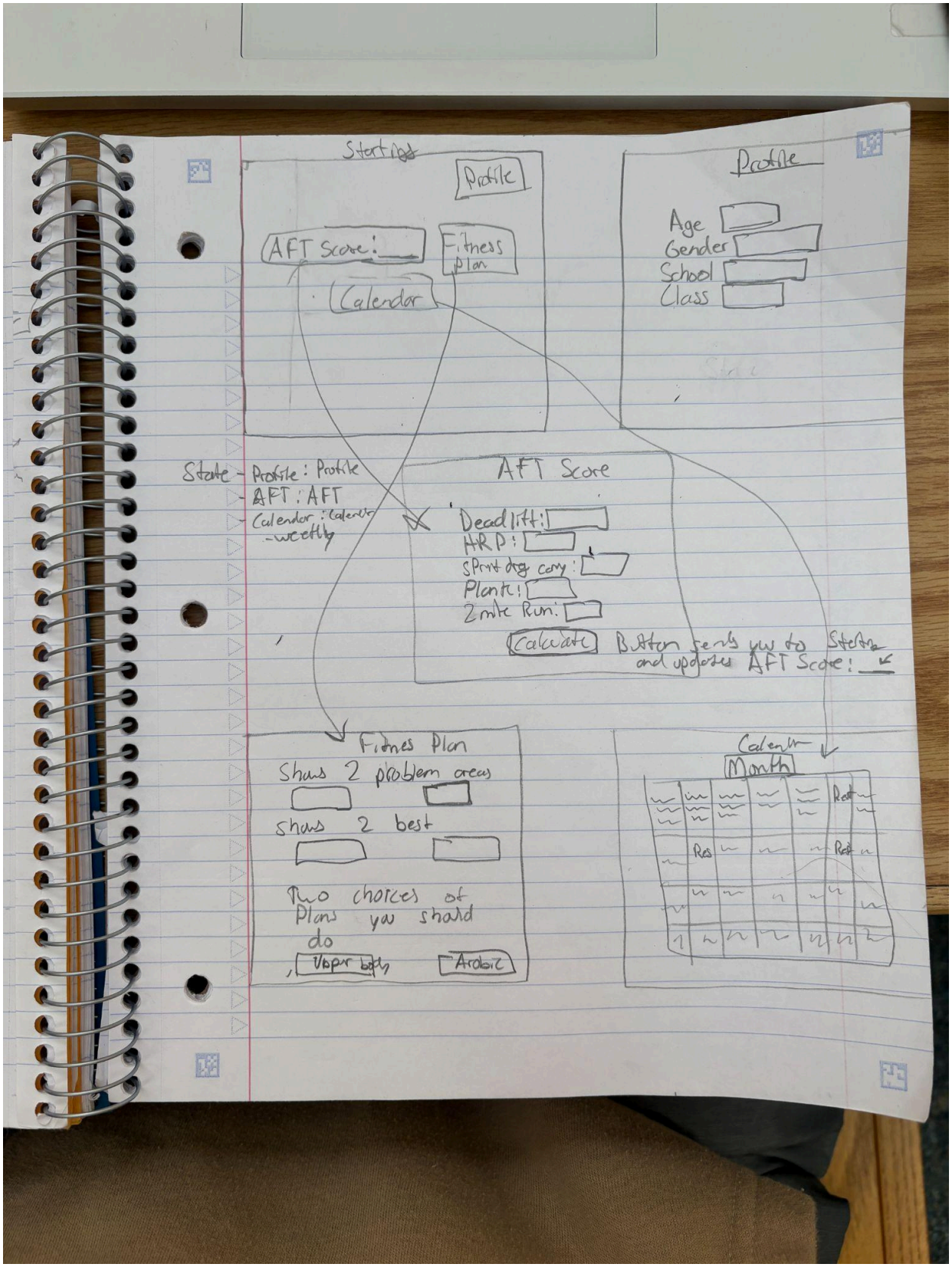
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


Start
Profile
AFT Score:
Fitness plan
Calendar
Profile
Age
Gender
School
Class
State - Profile: Profile
- AFT: AFT
- Calendar: Calendar
-weekly
AFT Score
Deadlift:
HRP:
Sprint drag carry:
Plank:
2 mile Run:
Calculate
Button sends you to Start and updates AFT Score:
Fitnes Plan
Shows 2 problem areas
shows 2 best
Two choices of Plans you should do
Upper body
Aerobic
Calendar
Month

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**This will help individuals in the army track their fitness progress and create a scheduled plan for their workouts**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	The state Dataclass will have Profile- Profile(DataClass) AFT-AFT(Dataclass) Calendar-2dList Fitness Plan - str
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Profile- Age:int Gender:str School:str Grade:int Ranger_Challenge:bool All of them will be modified by the user to work the whole program. The first 2 attributes(Age & Gender) are for the AFT calculation. The School and Ranger_challenge is used for the fitness plan portion and calendar. AFT-

	Deadlift:int HRP:int Sprint drag carry: str(#:##) this is minutes and seconds Plank: str(#:##) this is minutes and seconds 2 Mile Run: str(##:##) minutes and seconds For aft I will need to convert the imputed time values into second I will also need to convert all of these numbers into an aft score 0-500.
Routes (minimum 7) The names of your route functions	Home Page AFT score -Calculate Profile Fitness Plan -Choice of fitness plan Calendar
Pages (minimum 5) A brief description of the pages in your site	Home page- Place where you can then go into the specific pages Profile- Page that lets you input your specific gender,age,school, class, and if you do ranger challenge AFT Score- Lets you calculate your aft score by inputting you specific marks on each test Fitness Plan- shows you your weakest areas and strongest areas then recommends what your specific fitness plan you should do. Ie if your running and sprint drag carry is poor then it will recommend a aerobic workout and if you struggle for pushups and

	plank will recommend upper body
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	Home page has 4 buttons to move you to a specific page On profile page will be 5 text boxes to input your data On AFt Score will be 5 text boxes to input marks and 1 button to calculate you score On fitness plan one button to choose your specific plan
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	If statements will be used mostly in the profile section for the aft scores. The scores are all bassed off of gender and age so if you input you are a male who is 18 then the scores will calculate different than if you are a woman who is 30. If statement for when you select your fitness it will then change how your calendar will look because if you choose upper body it will have more upper body workouts each week versus if you choose aerobic it will give you more running workouts
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	The loops will be used for looping over each aft score that you input to do a sum for loop to add them all up. It is meaningful to the site because a large portion of the site is all about the aft score and if I cant add up the score my site will be less useful.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I did not use any LLM in this assignment

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.